INCH-POUND
MIL-PRF-32306
5 September 2008

PERFORMANCE SPECIFICATION

HOSE ASSEMBLY, PREFORMED FLEXIBLE, VIBRATION AND MOTION ISOLATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for a preformed one-piece construction flex hose assembly (continuous hose assembly) with captured end-flanges that are free to rotate (spin). Pressure capabilities of any hose meeting the first article tests prescribed herein are acceptable, however, the nominal pressure capability, which will vary by hose size, is expected to range around 150 pounds per square inch gauge (psig). Details concerning vibration dampening characteristics are beyond the scope of this document (see 6.1).

1.2 Classification. Preformed hose assemblies are of the type, size, and shape specified in the solicitation or contract (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-162	-	Hose, Rubber, Visual Inspection Guide for

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-2104	-	Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service
MIL-PRF-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-PRF-17672	-	Hydraulic Fluid, Petroleum, Inhibited
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-177 - Rubber Products, Terms for Visible Defects of

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN BUREAU OF SHIPPING (ABS)

ABS Rules for Building and Classing Naval Vessels (NVR) Part 5, Chapter 1, Section 1B - Auxiliary Machinery Systems, General Provisions, Metallic Piping and Installation

(Copies of this document are available from the American Bureau of Shipping, Government Operations Office, 1421 Prince Street, Suite 100, Alexandria, VA 22314 or online at www.eagle.org.)

ASME INTERNATIONAL

ASME B16.5 - Pipe Flanges and Flanged Fittings: NPS ½ through 24 (DoD adopted)

(Copies of this document are available from ASME International, 22 Law Drive, P.O. Box 2900, Fairfield, NJ 07007-2900 or online at www.asme.org.)

ASTM INTERNATIONAL

ASTM D380 - Standard Test Methods for Rubber Hose (DoD adopted)

ASTM D413 - Standard Test Methods for Rubber Property-Adhesion to Flexible Substrate (DoD adopted)

ASTM D1141 - Standard Practice for the Preparation of Substitute Ocean Water (DoD adopted)

(Copies of these documents are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

INTERNATIONAL MARITIME ORGANIZATION (IMO)

International Code for Application of Fire Test Procedures (FTP Code)

(Copies of this document are available by emailing publications-sales@imo.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material.

3.2.1 Hose assembly. The hose assembly shall consist of a tube, a reinforcement, and a cover, with integral flanges on each end that can be rotated as necessary for bolt hole alignment. The tube shall interface with the flange in such a way to provide an integral face seal.

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3.2.1.1 Tube. The tube shall be fabricated from synthetic rubber or equivalent, shall have a smooth bore, and shall be free from pitting, cuts, dirt, foreign material, mandrel lubricants, or other defects that would affect the performance of the hose. The tube shall be integral with and continuous from sealing surface on one flange to the other.

3.2.1.1.1 Flange sealing surface. The sealing surfaces on both ends of the hose assembly shall be integral with the tube construction. Additional gaskets or seals shall not be required. The sealing surface shall be proven sufficiently durable to survive several assembly/disassembly evolutions over the life of the hose assembly. The sealing surface of the hose shall show no evidence of material transfer to either the assembly flange or to the mating system flange during any of the acceptance tests listed in this specification.

3.2.1.2 Reinforcement. To ensure the durability of the hose, the reinforcement shall consist either of high strength carbon steel, stainless steel, high quality polyester or aramid fibers, or a combination thereof or equivalent. If a particular reinforcement material is desired, it shall be as specified (see 6.2). When polyester or aramid fibers are used, they shall have been processed and treated to obtain high strength and low creep properties and they shall be evenly and firmly braided, spirally wrapped, or for laminated fabric, layered in such a manner as to meet the requirements of 3.2.1.5, 3.3.2, 3.4, and 3.7. For braided and spiral wrapped reinforcement, there shall be no broken reinforcing fibers, nor shall any reinforcing fibers be omitted in any length of hose. At the discretion of the manufacturer, additional friction fabric layers or layers of rubber may be applied between reinforcements.

3.2.1.3 Cover. The cover shall be a low smoke and low toxicity composition of synthetic rubber compound to give maximum resistance to oxidation, weathering, abrasion, and light aging.

3.2.1.4 Flanges. Unless otherwise specified (see 6.2), flanges shall be designed to mate with ASME B16.5 flat-faced flanges. As specified (see 6.2), flanges shall be fabricated from metal or other material with consideration given to compatibility with normal shipboard environment and at least occasional contact with seawater.

3.2.1.4.1 Flange rotation. The end-flanges shall be free to rotate and be incorporated as an integral part of the hose assembly. The surface area of the seal on the flange face shall be as large as possible without interfering with the bolt holes.

3.2.1.5 Overall construction. The material and overall construction of the hose assembly shall be uniform in quality and free from defects that affect performance, serviceability, or appearance, such as delamination, lumps, kinks, splits, abrasions, scrapes, corroded surfaces, and impurities. FED-STD-162 shall be used for guidance.

3.2.2 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new.

3.3 Design.

3.3.1 Fabrication. Fabrication shall provide a one-piece hose assembly to meet the installation design requirements. Length and shape shall be as specified (see 6.2 and 6.3).

3.3.2 Concentricity. (see 6.5.1) Hose wall thickness measured in accordance with ASTM D380 shall not exceed the following values specified in table I.

TABLE I. Concentricity requirements.

Size		Concentricity (Inches)
Dash Number	Inches	
4	¼	0.060
>4≤32	≤2	0.100
>32	>2	0.125

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3.3.3 Compatibility. The hose assembly shall be designed for the following service fluid applications: seawater, fresh water, gray water, black water, water/glycol mixtures, aqueous film forming foam (AFFF), fuels, hydraulic fluids, lubricating oil, and compressed air.

3.4 Physical requirements.

3.4.1 Proof pressure. The hose assembly shall show no evidence of leakage, rupture, or deformation when subjected to the proof pressure test specified in 4.5.2.

3.4.2 Stability (dimensional change). When tested in accordance with 4.5.3, hoses shall not change in length more than plus 2 percent or minus 4 percent.

3.4.3 Aging. The hose assembly shall show no evidence of aging, such as cracking, separation of piles, or a deformation in the form of bulging or collapsing when subjected to the tests specified in 4.5.4.1 or 4.5.4.2 (high temperature). Neither shall the sealing surfaces show evidence of material degradation or pull-out of material on mating flange faces.

3.4.4 Hydraulic fluid or oil circulation (nominal and high temperature). There shall be no evidence of leakage through the hose or end connections when subjected to the hydraulic fluid or oil circulation test specified in 4.5.5.1 (nominal temperature) and 4.5.5.2 (high temperature).

3.4.5 Hydraulic impulse. The hose assembly shall show no evidence of failure or leakage when subjected to the hydraulic impulse tests specified in 4.5.6.1 or 4.5.6.2.

3.4.6 Burst pressure. Hoses shall be designed to withstand the burst pressure test as specified in 4.5.7 without indication of failure or leakage.

3.4.7 Vacuum. Without the use of an internal support device, there shall be no evidence of ply separation, collapse, or other degradation of the hose assembly when vacuum tested as specified in 4.5.8.

3.4.8 Cold temperature flexibility. The hose assembly shall be designed for application of -40±2 °F (-40±2 °C) temperatures.

3.4.9 Adhesion between plies. The adhesion between the tube, cover, and reinforcement shall be sufficient to withstand normal system operation within a shipboard environment when tested in accordance with 4.5.10.

3.5 Environmental conditions.

3.5.1 Temperature range. Hose assemblies shall comply with the specified requirements through a fluid temperature range of -40 °F (-40 °C) to 200 °F (93 °C) for all applications. Hose assemblies for use with both petroleum and phosphate ester based lubricants shall have a temperature range of -55 °F (-48 °C) to 300 °F (149 °C).

3.5.2 Fire resistance. The fire resistant requirements shall be those specified for a non-metallic flexible hose in NVR Part 5, Chapter 1, Section 1B, paragraph 3.5.1(e) titled “Fire resistant test.”

3.5.3 Flame spread. The hose assembly shall not sustain or propagate flames when tested in accordance with IMO Resolution A.653(16) of the FTP Code.

3.6 Identification/marking. Marking of finished product shall not deviate from that recommended below and shall provide required information in a clearly legible format.

3.6.1 Identification. The hose assembly shall have a permanent and visible embossed or inlaid lay-line along its entire length. The numerals and lettering shall be not less than $\frac{3}{16}$ inch high, and each set of markings shall be provided at two locations somewhere on the assembly. The two areas of marking shall contain, but not be limited to: (1) this performance spec number, (2) hose dash size (such as -16 for a 1-inch hose), (3) day/month/year of manufacture (dd/mm/yyyy), (4) manufacturer’s CAGE code, and (5) rated service pressure. If the assembly is too small to allow legible marking, attached identification tags may be used.

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3.7 Workmanship. The hose assembly shall be uniform in quality and material. Hose tube shall be smooth with no reinforcement visible and shall be free from dirt. Cover shall completely cover the reinforcement plies and shall be free of voids, blisters, cuts, nicks, and other defects that could impair the function and serviceability of the hose in its intended use.

3.8 Shelf-life. Shelf-life requirements shall be as specified (see 6.2 and 6.6).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.1.1 Inspection conditions. Unless otherwise specified herein, environmental conditions shall be ambient laboratory conditions.

4.2 First article inspection. First article inspection shall be performed on one complete hose assembly when a first article sampling is required (see 3.1). This inspection shall include the examination of 4.3 and 4.4 and the tests of 4.5 through 4.6.1 (see table II).

4.2.1 Sampling for first article testing. Sampling of hose assemblies shall be performed on three sample hose assemblies for each size. Failure of any sample shall be cause for rejection of that size unless otherwise specified herein.

4.3 Conformance inspection. Conformance inspection shall be accomplished on every hose and shall consist of the examinations and tests as specified in table II.

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TABLE II. First article tests and conformance inspections.

Examination/Test	Requirement	Test	First Article	Conformance
Visual	3.2-3.3.2, 3.4-3.4.7, and 3.6-3.7	4.4	X	X
Compatibility	3.3.3	4.5.1	X	X
Proof Pressure	3.4.1	4.5.2	X	X
Stability (Dimensional Change)	3.4.2	4.5.3	X	X
Aging	3.4.3	4.5.4.1	X	
Aging (High Temperature)	3.4.3	4.5.4.2	X	
Hydraulic Ambient Temperature Circulation	3.4.4	4.5.5.1	X	
Hydraulic Hot Fluid or Oil Circulation (Wire Reinforced, High Temperature)	3.4.4	4.5.5.2	X	
Hydraulic Impulse	3.4.5	4.5.6.1	X	
Retest & Averaging Impulse	3.4.5	4.5.6.2	X	
Burst Pressure	3.4.6	4.5.7	X	
Vacuum	3.4.7	4.5.8	X	
Cold Temperature Flexibility	3.4.8	4.5.9	X	
Adhesion	3.4.9	4.5.10	X	
Temperature Range	3.5.1	4.5.4.2	X	
Fire Resistance	3.5.2	4.6	X	
Flame Spread	3.5.3	4.6.1	X	

4.4 Visual examination. A visual examination shall be performed on each hose assembly to insure compliance with the requirements specified in table II. This examination shall encompass all visual examinations including compliance verification of system design and basic installation requirements. Noncompliance with any specified requirements or presence of one or more defects listed in table III shall constitute cause for rejection. Evaluation of defects shall be in accordance with FED-STD-162. Terms for describing the visible defects shall be as defined in MIL-STD-177.

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TABLE III. Classification of defects.

Category	Defects
1	Hose assembly does not conform to dimensions and tolerances specified
2	Reinforcement not as specified
3	Hose assembly size not as specified
4	Length not as specified
5	Marking missing or not as specified
6	Missing braid or spiral ply
7	Crack in cover or tube
8	Defective flanges
101	Tube not as specified
102	Missing or broken ply wire
103	Color not as specified
201	Depressed area of hose (exceeds outside diameter minimum tolerance)
202	Depressed area of hose tube (exceeds inside diameter tolerance
203	Presence of foreign material imbedded in hose assembly

4.5 Tests.

4.5.1 Compatibility. Evidence to support claims of compatibility shall be provided by either: (1) standard industry practice, or (2) specific material testing results acceptable to NAVSEA.

4.5.2 Proof pressure. The proof pressure test of the hose assembly shall be conducted in accordance with ASTM D380 except that the samples shall be in accordance with 4.2.1 for first article tests and for conformance tests. The proof pressure shall be 200 percent of the maximum working pressure and shall be held for not less than 1 minute or greater than 5 minutes. The pressure shall be increased at a uniform rate. Hose assemblies which pass during this test are acceptable for use. All high pressure hose assemblies shall follow ASTM D380 for the rate of pressure increase. Evidence of leakage, rupture, or deformation of the hose assembly shall constitute failure of this test. Nonconformance to 3.4.1 shall constitute failure of this test. Each hose assembly passing this test shall have a tag affixed with the following information: test facility, date, working pressure, tested pressure, and contract reference number. As specified (see 6.2), a paper certification shall be included with the hose assembly when shipped and sent to the address identified.

4.5.3 Stability (dimensional change). After successful completion of the proof test, the hose assembly samples shall be measured as specified in ASTM D380 except that they shall be marked at least 0.50 inch from each end fitting and the distance measured between the marks along the lay-line. The maximum working pressure shall then be applied and the distance between the marks measured at that pressure. Nonconformance to 3.4.2 shall constitute failure of this test.

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4.5.4 Aging of samples.

4.5.4.1 Aging of samples. The assembly shall be completely filled and pressurized to working pressure with oil or hydraulic fluid in accordance with MIL-PRF-2104, MIL-PRF-17331, MIL-PRF-17672, or MIL-PRF-83282 and held at this pressure while immersed in a non-pressurized, closed-type container or a reflux-type condenser to prevent distillation of the volatile matter in the fluid. Each assembly shall be aged by soaking it in the operating fluid that the hose would be exposed to during its service life. If that fluid is not specified in the solicitation or contract, each assembly shall be aged by being subjected to a temperature of 160±5 °F, first in air, then immersed in synthetic seawater that meets or exceeds ASTM D1141 for the times specified below:

a. 76 hours ±0.25 hour in air.

b. 46 hours ±0.25 hour in synthetic seawater.

Provision shall be made to bleed off the excess pressure caused by thermal expansion of the pressurized fluid. Nonconformance to 3.4.3 shall constitute failure of this test.

4.5.4.2 Aging of samples for high temperature applications. The sample assembly shall be filled with oil or hydraulic fluid conforming to MIL-PRF-2104, MIL-PRF-17331, MIL-PRF-17672, or MIL-PRF-83282 and placed in an oven at a temperature of 300±5 °F for 168 hours. During aging, all air must be excluded from the hose bore and provisions made to prevent pressure build-up due to expansion of the oil or hydraulic fluid. At the completion of the 168-hour period, the assembly shall be permitted to cool to room temperature and the oil or hydraulic fluid removed. The assembly shall be examined for a collapsed or an excessively bulged inner tube. There shall be no evidence of collapse of the inner tube or excessive bulging of the inner tube. Upon completion of the oil or hydraulic fluid aging test, the assembly shall be drained and placed in an oven at 300±5 °F for a period of 168 hours. At the completion of the 168-hour period, the assembly shall be permitted to cool to room temperature. The assembly shall then be examined for any evidence of degradation including inner tube cracks and collapse or sealing face damage. Nonconformance to 3.4.3 shall constitute failure of this test.

4.5.5 Hydraulic fluid or oil circulation.

4.5.5.1 Hydraulic fluid or oil circulation - nominal temperature. The assembly shall be filled with oil or hydraulic fluid in accordance with MIL-PRF-2104, MIL-PRF-17331, MIL-PRF-17672, or MIL-PRF-83282. The hose assembly shall then be pressurized from 75 to 100 psig. The fluid temperature shall be raised from ambient temperature to 180±10 °F and the fluid circulated through the hose assembly at a flow rate of not less than 3 gallons per minute. The ambient temperature shall be maintained at 75±15 °F. At least once during every 24 hours of testing, the pressure shall be increased to the design working pressure of the hose assembly and maintained for not less than 5 minutes. The flow shall be continued for not less than 200 hours. At the completion of this test, pressure shall be increased to the hose assembly's design working pressure and held for not less than 5 minutes. Nonconformance to 3.4.4 shall constitute failure of this test.

4.5.5.2 Hot hydraulic fluid or oil circulation - high temperature. For wire reinforced high temperature applications, the hose assembly shall be filed with oil or hydraulic fluid in accordance with MIL-PRF-2104, MIL-PRF-17331, MIL-PRF-17672, or MIL-PRF-83282. Assemblies shall be installed and tested on a flex and circulation stand as shown on figure 1. Oil or hydraulic fluid maintained at 300±5 °F shall be circulated through assemblies at 60 pounds per square inch (psi) while the assemblies are flexed 3.5 inches in each direction at 12±4 cycles per minute. The hose assembly shall be subjected to 400 hours of oil or hydraulic fluid circulation and flexing without evidence of leakage or other malfunction. The oil or hydraulic fluid shall be changed after 200 hours of circulation. Nonconformance to 3.4.4 shall constitute failure of this test.

4.5.6 Hydraulic impulse. Preformed hose assemblies shall be installed in the impulse test stand, figure 1, with the ends constrained between manifolds. The flexing manifold shall be adjusted such that at the center of its travel cycle no hose distortion is imposed.

4.5.6.1 Hydraulic impulse. Three hose assemblies of each size shall be subjected to an impulse rate from 30 to 80 cycles per minute with the hydraulic fluid at a temperature of 125±5 °F for 150,000 cycles. Test fluid may flow through the test sample in order to maintain temperature. Each pressure cycle shall attain 120 percent of the maximum working pressure and shall not fall below the applicable working pressure before the hose is depressurized (see figure 2). Nonconformance to 3.4.5 shall constitute failure of this test.

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4.5.6.2 Retest and averaging impulse. In the event of a hose assembly failure during impulse testing, an additional hose assembly shall be tested and the results of all assemblies of the same size tested shall be averaged. The conditions for testing an additional hose assembly shall be as follows:

a. The failed hose assembly shall have reached not less than 95 percent of the required number of impulse cycles.

b. For the purpose of averaging the number of cycles, the maximum number of cycles that can be used from any single tested hose assembly is 110 percent of the required number of cycles, or the actual number of cycles at the time of failure if failure occurs between 100 and 110 percent of the required number of cycles.

Nonconformance to 3.4.5 shall constitute failure of this test.

4.5.7 Burst pressure. The burst pressure test of the hose assembly shall be conducted in accordance with ASTM D380 except that the samples shall be in accordance with 4.2.1. The pressure shall be increased at a uniform rate to burst pressure of 4 times the maximum working pressure of hose assembly. Evidence of leakage, rupture, or deformation of the hose assembly at or below test pressure shall constitute failure of this test. Hose assemblies used for the burst pressure test shall not be acceptable for further use. Nonconformance to 3.4.6 shall constitute failure of this test.

4.5.8 Vacuum. The specified hose assembly shall be additionally tested under a vacuum not less than 28 inches of mercury for not less than 30 seconds. Nonconformance to 3.4.7 shall constitute failure of this test.

4.5.9 Cold temperature flexibility. Cold temperature flexibility testing on the tube and cover as described in ASTM D380 shall be performed. A separate piece of hose having the same specifications shall be used as the designated sample tube and cover assembly and shall be tested at -40±2 °F (-40±2 °C).

4.5.10 Adhesion. Adhesion of the tube and cover to the reinforcement, and plies of reinforcement to each other, shall be determined in accordance with ASTM D413, Static Mass Method, using Type A strip specimen procedure, except that the required minimum weight shall be supported by the test specimen for 3 minutes without causing a separation greater than 1 inch. Three specimens of each interface shall be tested. A separation rate greater than l inch in 3 minutes shall constitute failure of this test.

4.6 Fire resistance. The fire testing shall be conducted in accordance with the test method for fire endurance testing of water filled plastic piping described in Appendix 2 of IMO Resolution A.753(18) of the FTP Code. Fire resistance of hose assemblies shall meet the requirements for flexible and expansion joints as specified in NVR Part 5, Chapter 1, Section 1B.

4.6.1 Flame spread. Materials/systems shall meet the surface flammability criteria of bulkhead, wall, and ceiling linings as listed in IMO Resolution A.653(16) of the FTP Code.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The intended use is for U.S. Navy Surface Ships. The hose products covered under this specification are for shipboard machinery motion and vibration isolation in new applications and for back-fit into existing installations. Prefabricated one-piece construction dogleg flex hose assemblies compensate for relative motion or excessive vibration between equipment and piping. Preformed hose assemblies in compliance with this specification are intended to convey fluids to and from equipment where motion between connections must be accommodated. Properly designed hose assemblies handle vibration as well as excursions from shock without damage. It is expected that these hose assemblies will be suitable for a variety of fluid applications. Guidance for use of preformed hose assemblies for their noise attenuating properties is provided in S6430-AE-TED-010.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Hose assembly type, size, and shape (see 1.2).

c. Specific issue of individual documents referenced (see 2.2.1 and 2.3).

d. When first article testing is required; either initial requirement, or based on change in application, design, material, or manufacturer (see 3.1).

e. Reinforcement material (see 3.2.1.2).

f. Flange type if other than flat-faced (see 3.2.1.4).

g. Type of metal or material if other than metal (see 3.2.1.4).

h. Hose leg lengths and included angle of hose assembly (see 3.3.1 and 6.3).

i. Documentation required for certification of hydrostatic testing and where to send the certifications (see 3.4.1 and 4.5.2).

j. Shelf-life requirements (see 3.8 and 6.6).

k. Packaging requirements (see 5.1).

l. Material safety data sheets (see 6.4).

6.3 Hose dimensions. The following guidance should be used for determining overall length, each leg length, angle, and elbow radius. It is expected that the majority of hose assemblies will have a 90 degree angle between the legs, but preformed hose construction is not limited to standard angles. Therefore the desired hose assembly configuration should be fully described (see 6.3.1 and 6.3.2). As directed in S6430-AE-TED-010, (1) straight hose configurations are to be avoided because they lack sound attenuating characteristics, and (2) the best way to determine the actual hose length for new hose installations is to template the actual shipboard conditions. From the template, measure the overall dimension of the hose assembly.

6.3.1 Dimensions of dogleg shaped hose assemblies. From the template, the following should be measured: (1) the overall length of hose between the sealing surfaces along the centerline, (2) each straight leg length, (3) the required angle between the legs, and (4) the desired elbow radius (see figure 3). A straight leg length is the distance from the sealing surface along the centerline of one leg to the intersection of the projected centerline of the other leg. The angle between the legs is the angle between centerlines at the projected point of intersection.

6.3.2 Dimensions of 180 degree U-bend hose assemblies. From the template, the following should be measured: (1) the overall length of hose between the sealing surfaces along the centerline, (2) the length of each straight leg, and (3) the distance between hose centerlines (see figure 4). Straight leg lengths are measured from the sealing face to the beginning of the curved portion of the hose. The distance between centerlines of the two straight legs will be equal to twice the radius of the curved portion.

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6.4 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract.

6.5 Definitions.

6.5.1 Concentricity. Concentricity is defined as the variance between maximum and minimum wall thickness measurement obtained from a series of measurements taken around the circumference of the hose but concentrated in the area visually suspected of being thick or thin.

6.6 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, *Shelf-life Management Manual*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.hq.dla.mil/.

6.7 Subject term (key word) listing.

90 degree

Dogleg

Flange

Preform

Reinforced rubber

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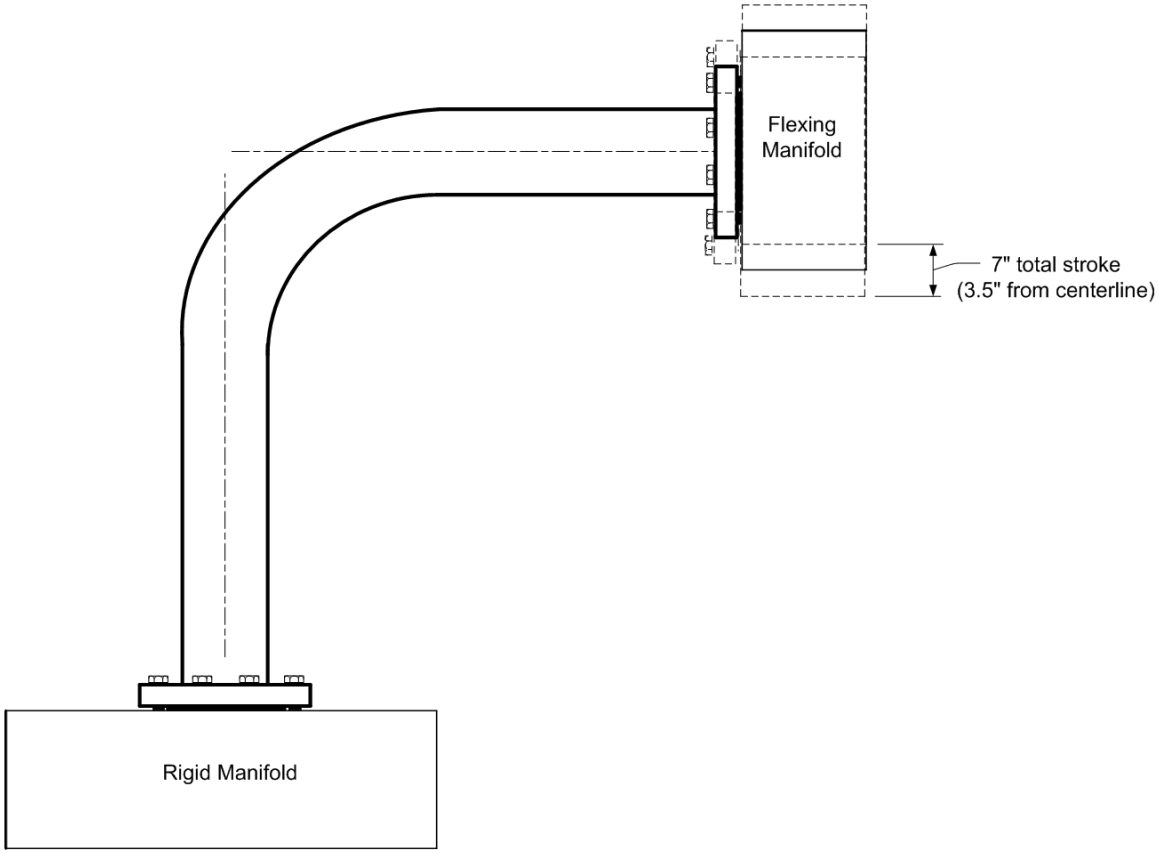
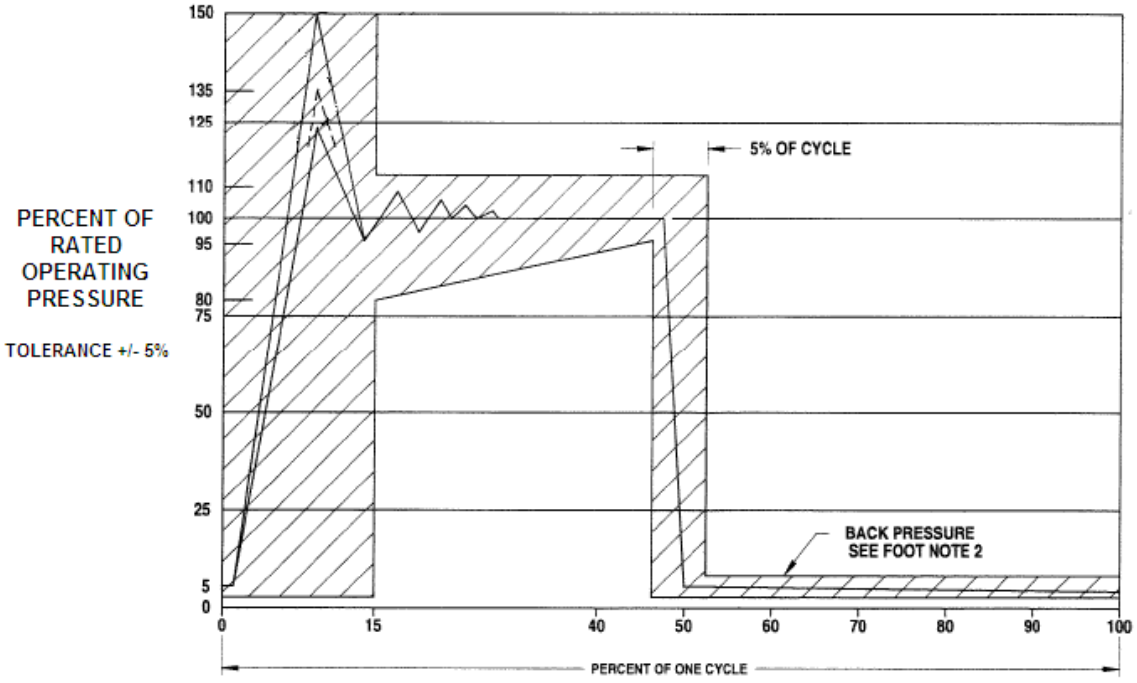


FIGURE 1. Flex and circulation test stand.

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NOTES:

1. This curve represents the desired pressure response for each cycle of the hydraulic impulse test specified in 4.5.6. Duration of each cycle should be between 1/30th and 1/80th of a minute. The actual pressure time cycle obtained when performing this test should fall within the shaded area.
2. The back pressure shall be 10±2 psi or, for higher pressure systems, in the range of 1 percent to 3 percent of the nominal pressure, whichever is greater.

FIGURE 2. Impulse test envelope.

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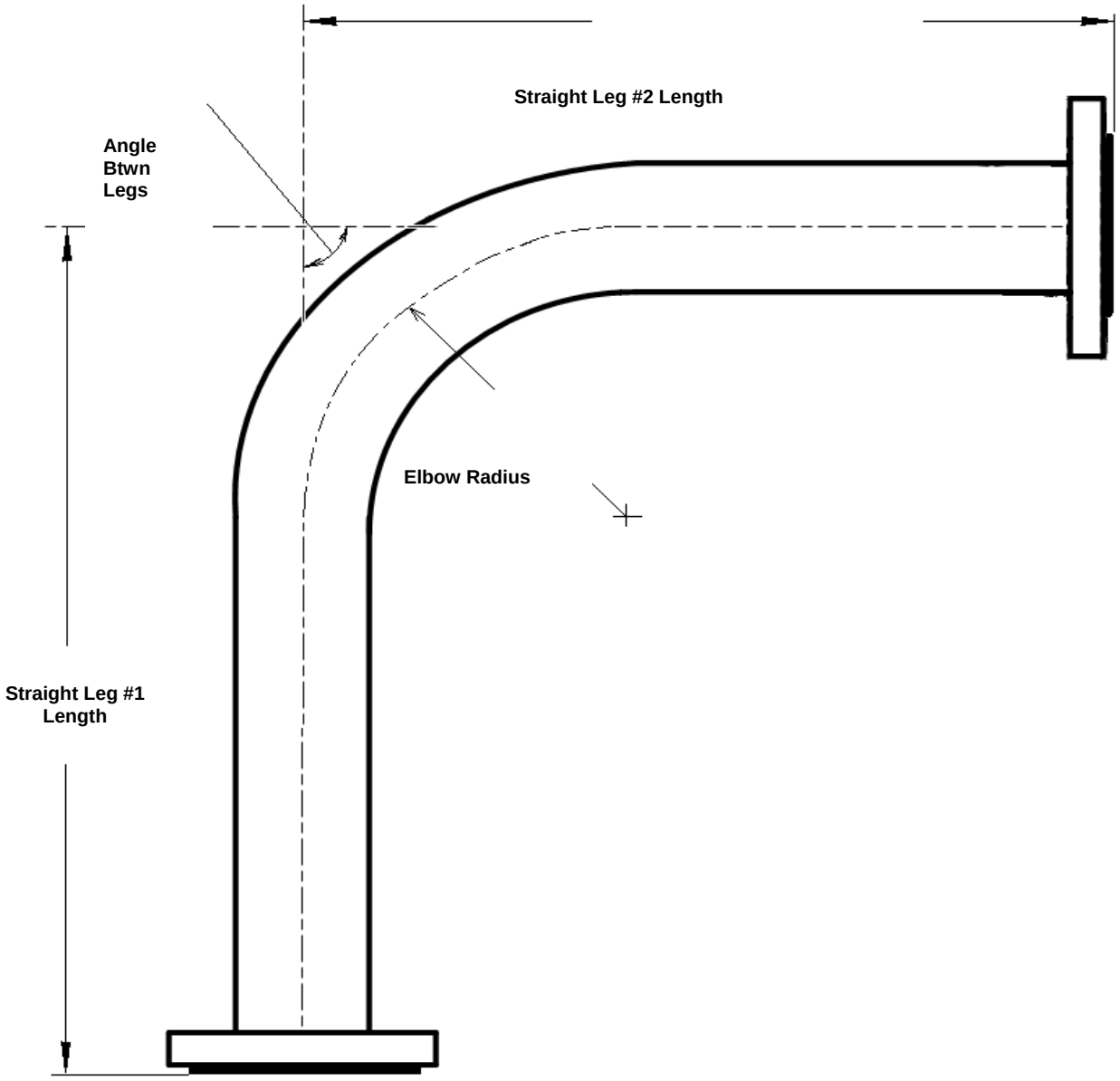


FIGURE 3. Measurement detail: dogleg<180 degrees.

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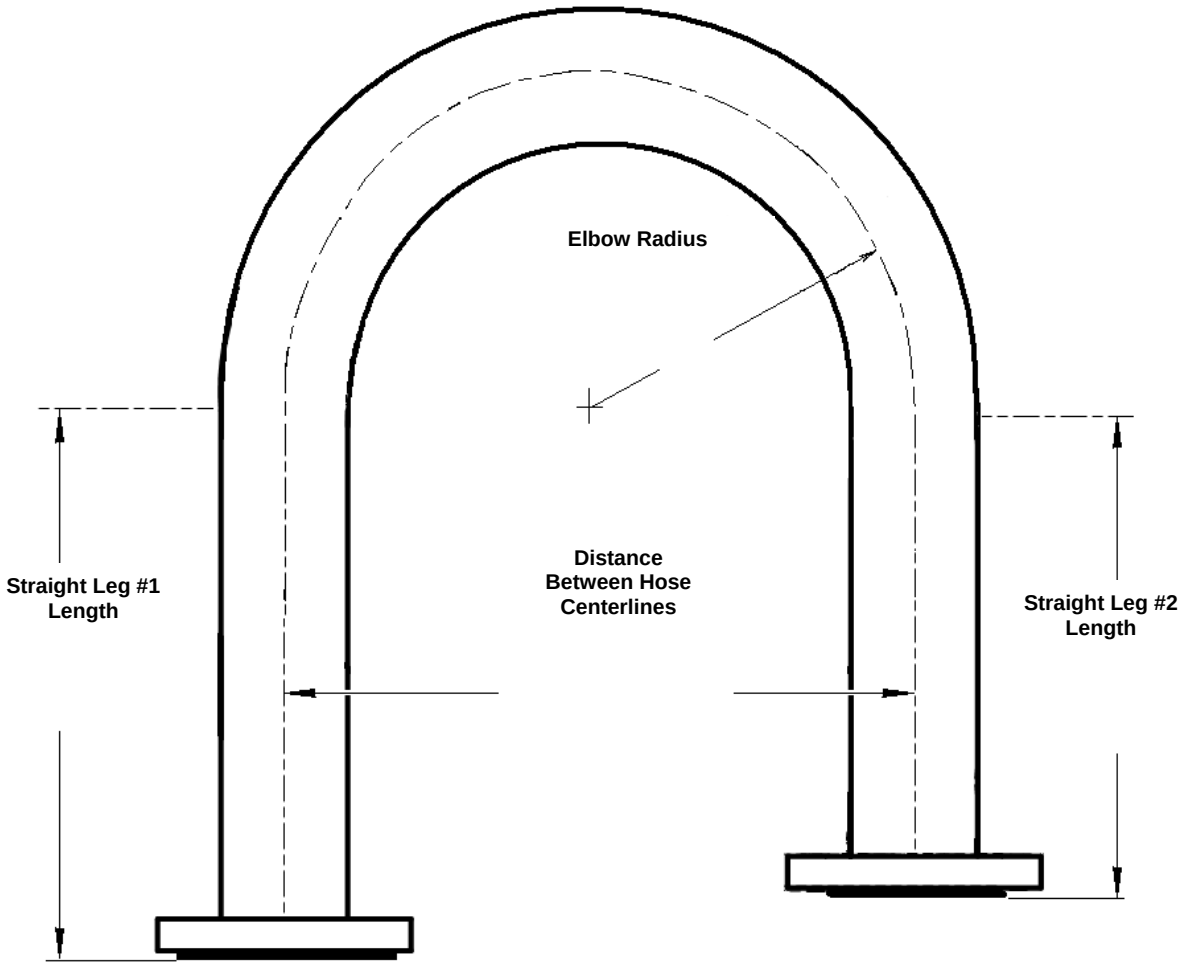


FIGURE 4. Measurement detail: 180 degree U-bend.

Custodians:
Navy – SH
Air Force – 99

Review activities:
Navy – AS, NP
Air Force – 03, 11, 71
DLA – CC
GSA – FSS

Preparing activity:
Navy – SH
(Project 4720-2008-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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